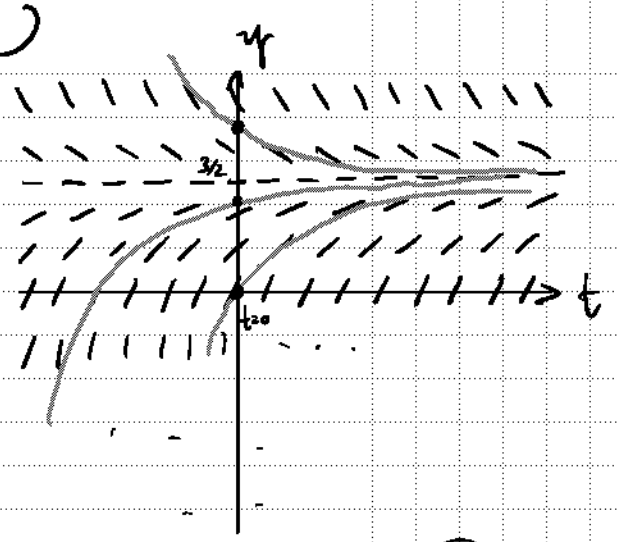
202 PS 1

In each of Problems 1 through 6, draw a direction field for the given differential equation. Based on the direction field, determine the behavior of y as $t \to \infty$. If this behavior depends on the initial value of y at $t = 0$, describe the dependency.

1. $y' = 3 - 2y$

2. $y' = 2y - 3$

(1)

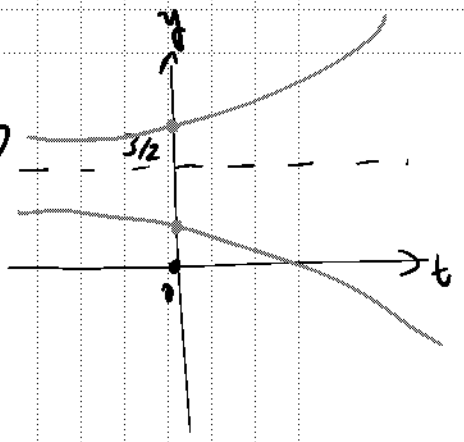


$y_0 = y(0) > 3/2$ as $t \to \infty$ $y \searrow 3/2$

$y(0) < 3/2$ as $t \to \infty$ $y \nearrow 3/2$

$y(0) = \frac{3}{2} \Rightarrow y = \frac{3}{2}$ as $t \to \infty$ $y = \frac{3}{2}$

(2)



$y(0) > \frac{3}{2}$ $y \to \infty$ as $t \to \infty$

$y(0) < \frac{3}{2}$ $y \to -\infty$ as $t \to \infty$

$y(0) = \frac{3}{2} \Rightarrow y \to \frac{3}{2}$ as $t \to \infty$

In each of Problems 1 through 6, determine the order of the given differential equation; also state whether the equation is linear or nonlinear.

PDE/ODE

lin/nonlin.

order

1. $t^2 \frac{d^2y}{dt^2} + t\frac{dy}{dt} + 2y = \sin t$

2. $(1 + y^2)\frac{d^2y}{dt^2} + t\frac{dy}{dt} + y = e^t$

3. $\frac{d^4y}{dt^4} + \frac{d^3y}{dt^3} + \frac{d^2y}{dt^2} + \frac{dy}{dt} + y = 1$

4. $\frac{dy}{dt} + ty^2 = 0$

5. $\frac{d^2y}{dt^2} + \sin(t + y) = \sin t$

6. $\frac{d^2y}{dt^2} + t\frac{dy}{dt} + (\cos^2 t)y = t^2$

1) 2nd order lin.
2) 2nd order nonlin.
3) 4th order lin.
4) 1st order nonlin.
5) 2nd order nonlin.
6) 3rd order lin.

ODE

"$\mathcal{L}[y]$ – lin?

In each of Problems 21 through 24, determine the order of the given partial differential equation; also state whether the equation is linear or nonlinear. Partial derivatives are denoted by subscripts.

21. $u_{xx} + u_{yy} + u_{zz} = 0$

22. $u_{xx} + u_{yy} + uu_x + uu_y + u = 0$

23. $u_{xxxx} + 2u_{xxyy} + u_{yyyy} = 0$

24. $u_t + uu_x = 1 + u_{xx}$

) PDE

21) 2nd
22) 2nd
23) 4th
24) 2nd

ord.

lin
nonlin
lin
nonlin.

$xy\,u_{xx} + 4u_y + tz\,u_{yy} = 0$

dep. var. u, indep. vars. (x, y, z, t)

$u(x, y, z, t)$

$= \sin u$ nonlin.

$u \cdot u_{xy}$ nonlin

u^2 nonlin.

e^t lin

e^u nonlin.

(c) Find the general solution of the given differential equation, and use it to determine how solutions behave as $t \to \infty$.

5. $y' - 2y = 3e^t$

10. $y' + 2y = te^{-2t}$, $y(1) = 0$

1st ord. lin $\Rightarrow$ μ integr. factor

5

$y' - 2y = 3e^t$

$\cdot \mu(t) \Rightarrow \int \frac{d}{dt}(\mu y)\,dt = \int 3e^t \cdot \mu(t)\,dt$

$I(t)$

$\cdot \mu(t)$

$\mu y' - 2\mu y = 3e^t \mu$

$= (\mu \cdot y)'$ if $= \mu' y + \mu y'$

$\mu y = I(t)$

$y = \frac{I(t)}{\mu(t)}$

$\mu' = -2\mu$

$$\frac{\mu'}{\mu} = -2$$

$$\int \frac{\mu'}{\mu}\,dt = -\int 2\,dt$$

$$\ln|\mu| = -2t + C$$

$$e^{\ln|\mu|} = e^{-2t+C}$$

$$|\mu| = e^{-2t}\cdot e^{C} \qquad C \in \mathbb{R}_{>0} \qquad \Rightarrow \quad \mu(t) = Ce^{-2t}$$

$$\underbrace{Ce^{-2t}y' - 2Ce^{-2t}y}_{(e^{-2t}y)'} = 3e^{t}\cdot Ce^{-2t}$$

$$\int \frac{d}{dt}(e^{-2t}y)\,dt = \int 3e^{-t}\,dt$$

$$e^{-2t}y = -3e^{-t} + C \qquad \cdot e^{2t}$$

$$y = -3e^{t} + C\cdot e^{2t} = \underbrace{e^{2t}}_{\to\infty}\left(\underbrace{-3e^{-t}}_{\to 0} + C\right)$$

(10) $y' + 2y = te^{-2t}$, $y(1) = 0$

$C > 0 \Rightarrow y \to \infty$

$C < 0 \Rightarrow y \to -\infty$

$$\mu y' + \underbrace{\mu'}_{2\mu} y = \mu t e^{-2t}$$

$$\mu' = 2\mu \Rightarrow \int \frac{\mu'}{\mu} = \int 2 \Rightarrow \ln|\mu| = 2t + C \Rightarrow |\mu| = e^{2t}\cdot C_{>0}$$

$$\mu = C\cdot e^{2t}$$

$$(\mu y)' = \mu y' + \mu' y = e^{2t} y' + 2e^{2t} y = e^{2t} t e^{-2t} = t$$

$(e^{2t} y)'$

$$\int (e^{2t} y)' \, dt = \int t \, dt$$

$$e^{2t} y = \frac{t^2}{2} + C \qquad (\cdot e^{-2t})$$

$$y = \frac{t^2 e^{-2t}}{2} + C e^{-2t} \longrightarrow \text{general soln.}$$

IV: $y(1) = \frac{1^2 \cdot e^{-2}}{2} + C \cdot e^{-2} = 0$

$$Ce^{-2} = -\frac{e^{-2}}{2} \qquad C = -\frac{1}{2}$$

Soln to the IVP: $y = \frac{t^2 e^{-2t}}{2} - \frac{e^{-2t}}{2} \longrightarrow 0$ as $t \to \infty$

16. $y' + (2/t)y = (\cos t)/t^2, \quad y(\pi) = 0, \quad t > 0$ 1st ord. lin.

$$\mu y' + \underbrace{\mu \frac{2}{t}}_{=\mu'} y = \mu \frac{\cos t}{t^2}$$

$$\mu' = \mu \frac{2}{t} \quad \Rightarrow \int \frac{\mu'}{\mu} dt = \int \frac{2}{t} dt$$

$$\ln|\mu| = 2\ln|t| + C$$

$$e^{\ln|\mu|} = e^{\ln|t|^2 + C} = e^{\ln t^2} \cdot e^C$$

$$\mu(t) = t^2 \cdot C_{\neq 0}$$

$$t^2 y' + \frac{2}{t} t^2 y = \frac{\cos t}{t^2} t^2$$

$$(\mu y)' = (t^2 y)' = \cos t$$

$$\int (t^2 y)' dt = \int \cos t \, dt$$

$$t^2 y = \sin t + C$$

$$y = \frac{\sin t}{t^2} + C t^{-2} \longrightarrow \text{gen. soln.}$$

IV: $y(\pi) = 0 \Rightarrow y(\pi) = \frac{\sin \pi}{\pi^2} + \frac{C}{\pi^2} = 0$

$$\sin \pi + C = 0 \quad \Rightarrow C = 0$$

$\sin \pi = 0$

$$y(t) = \frac{\sin t}{t^2}$$

30. Find the value of y_0 for which the solution of the initial value problem

$$y' - y = 1 + 3\sin t, \qquad y(0) = y_0$$

remains finite as $t \to \infty$.

$$\mu y' - \mu y = (1 + 3\sin t)\mu$$

$\mu' = -\mu$

$$\int \frac{\mu'}{\mu} = \int -1 \quad \Rightarrow \quad \ln|\mu| = -t + C$$

$$\mu = C e^{-t}$$

$$(e^{-t}y)' = (1 + 3\sin t)e^{-t}$$

$$\int (e^{-t}y)'\,dt = \int e^{-t}\,dt + 3\int e^{-t}\sin t\,dt \qquad (*)$$

$e^{-t}y$; $-e^{-t}$; by parts

$$\int \underbrace{e^{-t}}_{u}\ \underbrace{\sin t\,dt}_{dv} = -e^{-t}\cos t - \int \underbrace{\cos t}_{dv}\ \underbrace{e^{-t}}_{u}\,dt$$

$\int u\,dv = uv - \int v\,du$

$$= -e^{-t}\cos t - \left[e^{-t}\sin t + \int \sin t\, e^{-t}\,dt\right]$$

$$= -e^{-t}\cos t - e^{-t}\sin t - \int \sin t\, e^{-t}\,dt$$

$$\int e^{-t}\sin t\,dt = \frac{1}{2}\left(-e^{-t}\cos t - e^{-t}\sin t\right) + C$$

$(*)$ $\qquad e^{-t}y = -e^{-t} + \frac{3}{2}\left(-e^{-t}\cos t - e^{-t}\sin t\right) + C \qquad (\cdot e^{t})$

$$y = -1 + \frac{3}{2}(-\cos t - \sin t) + Ce^t \quad \rightarrow \text{gen. soln.}$$

IV: $y(0) = y_0 \rightarrow y(0) = -1 + \frac{3}{2}(-1-0) + Ce^0 = y_0$

$$-\frac{5}{2} + C = y_0$$

$$C = y_0 + \frac{5}{2}$$

Soln to IVP: $y = -1 + \frac{3}{2}(-\cos t - \sin t) + \left(y_0 + \frac{5}{2}\right)e^t$

$-1 \le \cos t \le 1$, $-1 \le \sin t \le 1$

$-2 \le -\cos t - \sin t \le 2$

$-3 \le -1 + \frac{3}{2}(-\cos t - \sin t) \le 3$

$-4 \le y \le 2$

$\left(y_0 + \frac{5}{2}\right)e^t \rightarrow \infty$

To kill the $\left(y_0 + \frac{5}{2}\right)e^t$ part, choose $y_0 = -\frac{5}{2}$

so that as $t \rightarrow \infty$ y remains finite ($-4 \le y(t) \le 2$ everywhere)

In each of Problems 1 through 8, solve the given differential equation.

1. $y' = x^2/y$ — 1st ord. nonlin. — 6. $xy' = (1 - y^2)^{1/2}$

$\frac{dy}{dx}\,dx$ $\qquad dy(x,t) = \frac{\partial y}{\partial x}\,dx + \frac{\partial y}{\partial t}\,dt$

Careful! here we lost the solutions $y=1$, $y=-1$

① $\int y\,dy = \int x^2 dx$

$\frac{y^2}{2} = \frac{x^3}{3} + c_1$

$y^2 = \frac{2x^3}{3} + c_1$ → implicit soln.

$y = \pm\sqrt{\frac{2x^3}{3} + c_1}$ → explicit soln.

⑥ $x\frac{dy}{dx} = \sqrt{1-y^2}$

$\int \frac{dy}{\sqrt{1-y^2}} = \int \frac{dx}{x}$

$\arcsin y = \ln|x| + c_1$

$y = \sin(\ln|x| + c_1)$

23. Solve the initial value problem

$$y' = 2y^2 + xy^2, \qquad y(0) = 1$$

and determine where the solution attains its minimum value.

$\int \frac{dy}{y^2} = \int (2 + x)\,dx$

$-\frac{1}{y} = 2x + \frac{x^2}{2} + c_1$

$y = \frac{1}{-2x - \frac{x^2}{2} + c_1} = -\frac{2}{4x + x^2 + c_1}$ → gen. soln.

IV: $y(0) = 1 \Rightarrow y(0) = -\frac{2}{c_1} = 1 \Rightarrow c = -2$

Soln to IVP: $y(x) = -\dfrac{2}{4x+x^2-2} = -2(4x+x^2-2)^{-1}$

Min. of $y(x)$:

$$y'(x) = +2(4x+x^2-2)^{-2}\cdot(4+2x) = 0 \Rightarrow 4+2x=0$$
$$x=-2$$

or = undef. $\Rightarrow x^2+4x-2=0$

$$(x+2)^2 = 6$$

$$x_{1,2} = -2 \pm \sqrt{6} \rightarrow \text{vertical asymptotes}$$

$$\lim_{x\to(-2+\sqrt{6})^+} y(x) = \lim_{x\to(-2+\sqrt{6})^+} -\frac{2}{\underbrace{x^2+4x-2}_{(x+2)^2-6}} = -\infty$$

$x > -2+\sqrt{6}$

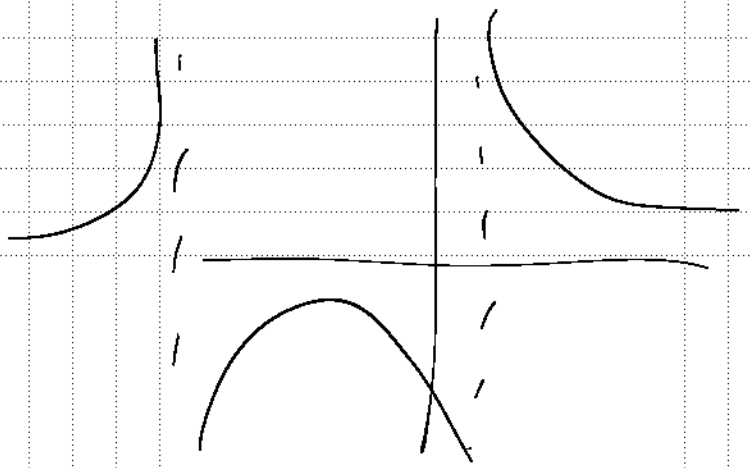
$x+2 > \sqrt{6}$

$(x+2)^2 > 6$

$(x+2)^2-6 > 0$

$y(x)$ doesn't have a minimum.

Sketch the graph to check



homogeneous

30. Consider the equation

$$\frac{dy}{dx} = \frac{y-4x}{x-y} \overset{?}{=} \frac{\lambda y - 4\lambda x}{\lambda x - \lambda y} \quad \text{(i)}$$

$$y' = F\left(\frac{y}{x}\right) = F\left(\frac{\lambda y}{\lambda x}\right)$$

(a) Show that Eq. (i) can be rewritten as

$$\frac{dy}{dx} = \frac{(y/x)-4}{1-(y/x)} = F\left(\frac{y}{x}\right) \quad \text{(ii)}$$

$$F(v) = \frac{v-4}{1-v}$$

thus Eq. (i) is homogeneous.

$$v = \frac{y}{x}$$

(b) Introduce a new dependent variable v so that $v = y/x$ or $y = xv(x)$. Express dy/dx in terms of x, v, and dv/dx.

(c) Replace y and dy/dx in Eq. (ii) by the expressions from part (b) that involve v and dv/dx. Show that the resulting differential equation is

$$v + x\frac{dv}{dx} = \frac{v-4}{1-v}$$

$$vx = y$$

$$v'x + v = y'$$

or

$$x\frac{dv}{dx} = \frac{v^2-4}{1-v} \quad \text{(iii)}$$

Observe that Eq. (iii) is separable.

(d) Solve Eq. (iii), obtaining v implicitly in terms of x.

(e) Find the solution of Eq. (i) by replacing v by y/x in the solution in part (d).

(d) $$\int \frac{1-v}{v^2-4}\,dv = \int \frac{dx}{x} = \ln|x| + c$$

$$\frac{1-v}{v^2-4} = \frac{1-v}{(v-2)(v+2)} = \frac{A}{v-2} + \frac{B}{v+2} = \frac{Av+2A+Bv-2B}{v^2-4}$$

$$\frac{A}{(v+2)} \quad \frac{B}{(v-2)}$$

$$\begin{cases} 1 = 2A - 2B \\ -1 = A + B \quad \times 2 \\ -2 = 2A + 2B \end{cases} \Rightarrow -1 = 4A \Rightarrow A = -\frac{1}{4},\ B = -\frac{3}{4}$$

$$\ln|x| + c_1 = \int \frac{1-v}{v^2-4}\,dv = \int -\frac{1}{4}\left(\frac{1}{v-2}\right) - \frac{3}{4}\left(\frac{1}{v+2}\right)dv$$

$$= -\frac{1}{4}\ln|v-2| - \frac{3}{4}\ln|v+2|$$

$$e^{\ln|x|+c} = e^{-\frac{1}{4}\ln|v-2| - \frac{3}{4}\ln|v+2|}$$

$$e^{\ln|x|}\cdot e^{c} = e^{\ln|v-2|^{-\frac{1}{4}} + \ln|v+2|^{-\frac{3}{4}}} = e^{\ln|v-2|^{-1/4}} \cdot e^{\ln|v+2|^{-3/4}}$$

$$|x| \cdot c_2 = |v-2|^{-1/4} \cdot |v+2|^{-3/4}$$

$v = \frac{y}{x}$ back substitute

$$C^4 \cdot x^4 = |v-2|^{-1} \cdot |v+2|^{-3}$$

$$x^4 = \frac{C}{\left|\frac{y}{x}-2\right| \cdot \left|\frac{y}{x}+2\right|^3}$$

$$= \frac{C}{\left|\frac{y-2x}{x}\right| \cdot \left|\frac{y+2x}{x}\right|^3}$$

$$x^4 = \frac{C x^4}{|y-2x| \cdot |y+2x|^3}$$

$$|y-2x| \cdot |y+2x|^3 = C$$

$$(y-2x)(y+2x)^3 = C$$

Ex & Un. for 1st ord. lin. ODE

Theorem 2.4.1 If the functions p and g are continuous on an open interval $I: \alpha < t < \beta$ containing the point $t = t_0$, then there exists a unique function $y = \phi(t)$ that satisfies the differential equation

$$1\ y' + p(t)y = g(t) \tag{1}$$

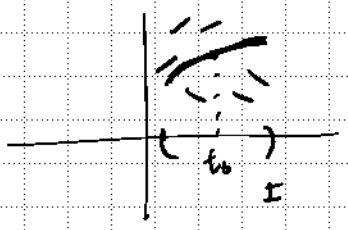
for each t in I, and that also satisfies the initial condition

$$y(t_0) = y_0, \tag{2}$$

where y_0 is an arbitrary prescribed initial value.

A soln y exists & unique on I → ① $t_0 \in I$

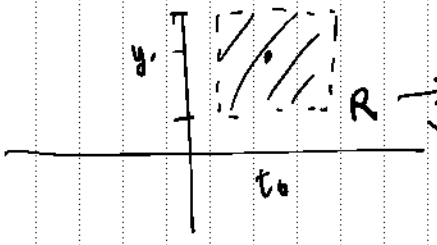
→ ② $p(t), g(t)$ cts on I



nonlin.

Theorem 2.4.2 Let the functions f and $\partial f/\partial y$ be continuous in some rectangle $\alpha < t < \beta$, $\gamma < y < \delta$ containing the point (t_0, y_0). Then, in some interval $t_0 - h < t < t_0 + h$ contained in $\alpha < t < \beta$, there is a unique solution $y = \phi(t)$ of the initial value problem

$$y' = f(t, y), \qquad y(t_0) = y_0. \tag{9}$$



→ ① f cts, $\frac{\partial f}{\partial y}$ cts

→ ② $(t_0, y_0) \in R$

In each of Problems 1 through 6, determine (without solving the problem) an interval in which the solution of the given initial value problem is certain to exist.

4. $(4 - t^2)y' + 2ty = 3t^2, \qquad y(-3) = 1$

① $-3 \in I$

② $1\ y' + \underbrace{\frac{2t}{4-t^2}}_{p(t)}\ y = \underbrace{\frac{3t^2}{4-t^2}}_{g(t)}$

$t = \pm 2$ discts

Continuity

f, g cts

- $f + g$ cts
- $f \cdot g$ cts
- f/g cts $\Leftrightarrow g \neq 0$

Then on $I = (-\infty, -2)$ a soln to the initial value prob. exists & it's unique.

Bernoulli Equations. Sometimes it is possible to solve a nonlinear equation by making a change of the dependent variable that converts it into a linear equation. The most important such equation has the form

$$y' + p(t)y = q(t)y^n,$$

and is called a Bernoulli equation after Jakob Bernoulli. Problems 27 through 31 deal with equations of this type.

27. (a) Solve Bernoulli's equation when $n = 0$; when $n = 1$.

(b) Show that if $n \neq 0, 1$, then the substitution $v = y^{1-n}$ reduces Bernoulli's equation to a linear equation. This method of solution was found by Leibniz in 1696.

(a) $y' + p(t)y = q(t)$ $(n=0)$

$y' + p(t)y = q(t)y$ $(n=1)$

$y' + (p(t) - q(t))y = 0$

(b) $n \geq 2$: $y' + p(t)y = q(t)y^n$ $(\cdot y^{-n})$

$v(t) = y^{1-n}(t)$

$v' = (1-n)y^{-n}(t)y'(t)$

$\frac{v'}{(1-n)} = y^{-n}(t)y'(t)$

$y' \cdot y^{-n} + p(t)y^{1-n} = q(t)$

$\frac{v'}{1-n}$ v

$(v = y^{1-n})$ $\Rightarrow$ $\frac{v'}{1-n} + p(t)v = q(t)$

$v' + p(t)(1-n)v = q(t)(1-n)$ $\rightarrow$ 1st ord. Eln.

use integrating factor

In each of Problems 28 through 31, the given equation is a Bernoulli equation. In each case solve it by using the substitution mentioned in Problem 27(b).

28. $t^2y' + 2ty - y^3 = 0, \quad t > 0$

$t^2y' + 2ty = y^3$

$y' + \frac{2}{t}y = t^{-2}y^3$ $\Rightarrow$ Bernoulli

$p(t)$ $q(t)$ $n=3$

$v = y^{-2}$

$v' = -2y^{-3} \cdot y'$

$y' \cdot y^{-3} + \frac{2}{t} y^{-2} = t^{-2}$

$\frac{v'}{-2}$ $\quad$ v

$\Rightarrow \quad \frac{v'}{-2} + \frac{2}{t} v = t^{-2} \qquad (\cdot(-2))$

$\mu v' - \frac{4}{t} \mu v = -2t^{-2}\mu \quad \longrightarrow$ 1st ord. lin.

$\mu' = -\frac{4}{t}\mu$

$\frac{\mu'}{\mu} = -\frac{4}{t}$

$c + \ln|\mu| = -4\ln|t| = \ln|t|^{-4} \quad \overset{\text{exp.}}{\Rightarrow} \quad \mu = ct^{-4}$

$t^{-4} v' - \frac{4}{t} t^{-4} v = -2t^{-2} \cdot t^{-4}$

$(\mu v)'$

$\int (t^{-4} \cdot v)' \, dt = \int -2t^{-6} \, dt$

$t^{-4} \cdot v = \frac{2}{5} t^{-5} + c \qquad (\cdot t^4)$

$v = \frac{2}{5} t^{-1} + c \cdot t^4 \qquad v = y^{-2}$

$\frac{1}{y^2} = y^{-2} = \frac{2}{5} t^{-1} + ct^4 = \frac{2}{5t} + ct^4 = \frac{2 + ct^5}{5t}$

$\frac{5t}{2 + ct^5} = y^2 \quad \longrightarrow$ Implicit sol.

$\pm \sqrt{\frac{5t}{2 + ct^5}} = y \quad \longrightarrow$ expl.